Nile Sun Power Limited
Solar EPC · Kampala, Uganda



COMMERCIAL PROPOSAL

Rooftop Solar PV System

Kampala Marriott Hotel · Mezzanine Roof

185.4 kWp | 299 × 620 Wp | 2 × 80 kW inverters | Zero export

PREPARED FOR
Kampala Marriott Hotel
Kampala, Uganda

PREPARED BY
Nile Sun Power Limited
Ref. Q-KMH-PV-2026-001 · 05 November 2026

Contents

03 Executive Summary 3
04 About Nile Sun Power 4
05 The Site 5
06 Proposed System 6
07 Energy Generation 7
08 Solar Energy Profile 8
09 Savings 9
10 Return on Investment 10
11 Financial Analysis 11
12 Cash Flow 12
13 Tariff Sensitivity 13
14 What's Included & Warranties 14
15 Commercial Quotation 15
16 Timeline & Contact 16

Executive Summary

Nile Sun Power Limited proposes a **185.4 kWp grid-tied rooftop solar PV system** on the mezzanine roof of Kampala Marriott Hotel. All solar energy is used on site (zero export, no battery), directly reducing the hotel's daytime electricity purchases from the grid.

185.4 kWp
solar capacity
299 × 620 Wp panels

274,362
kWh generated
in year 1

USD 19,212
savings in year 1

6.1 yrs
simple payback

16.6 %
project IRR

USD 71,221
NPV at 10 %

Investment: USD 121,115 turnkey EPC (excl. VAT) · **25-year net savings:** USD 569,043 · **Cost of solar energy:** USD 0.052/kWh vs grid USD 0.070/kWh

- ✔ Panels placed strictly within the architect's designated solar zone (drawing D18102-DD-A-104).
- ✔ Ballasted aluminium structure at 10° tilt – no roof penetrations.
- ✔ Zero-export control keeps all solar power inside the hotel.
- ✔ Smart monitoring of generation and savings from day one.

About Nile Sun Power

Nile Sun Power Limited is a solar energy company based in Kampala, Uganda, delivering turnkey solar PV systems for commercial and industrial customers – from site assessment and engineering through supply, installation, commissioning and after-sales support.

Engineering
site survey, layout, electrical design

Supply
Tier-1 panels, inverters & structures

Installation
certified team, safe rooftop works

Commissioning
testing & grid synchronisation

Monitoring
online performance tracking

Support
warranty & after-sales service

Our promise: a reliable, bankable solar system that lowers your energy costs and carbon footprint for 25+ years.

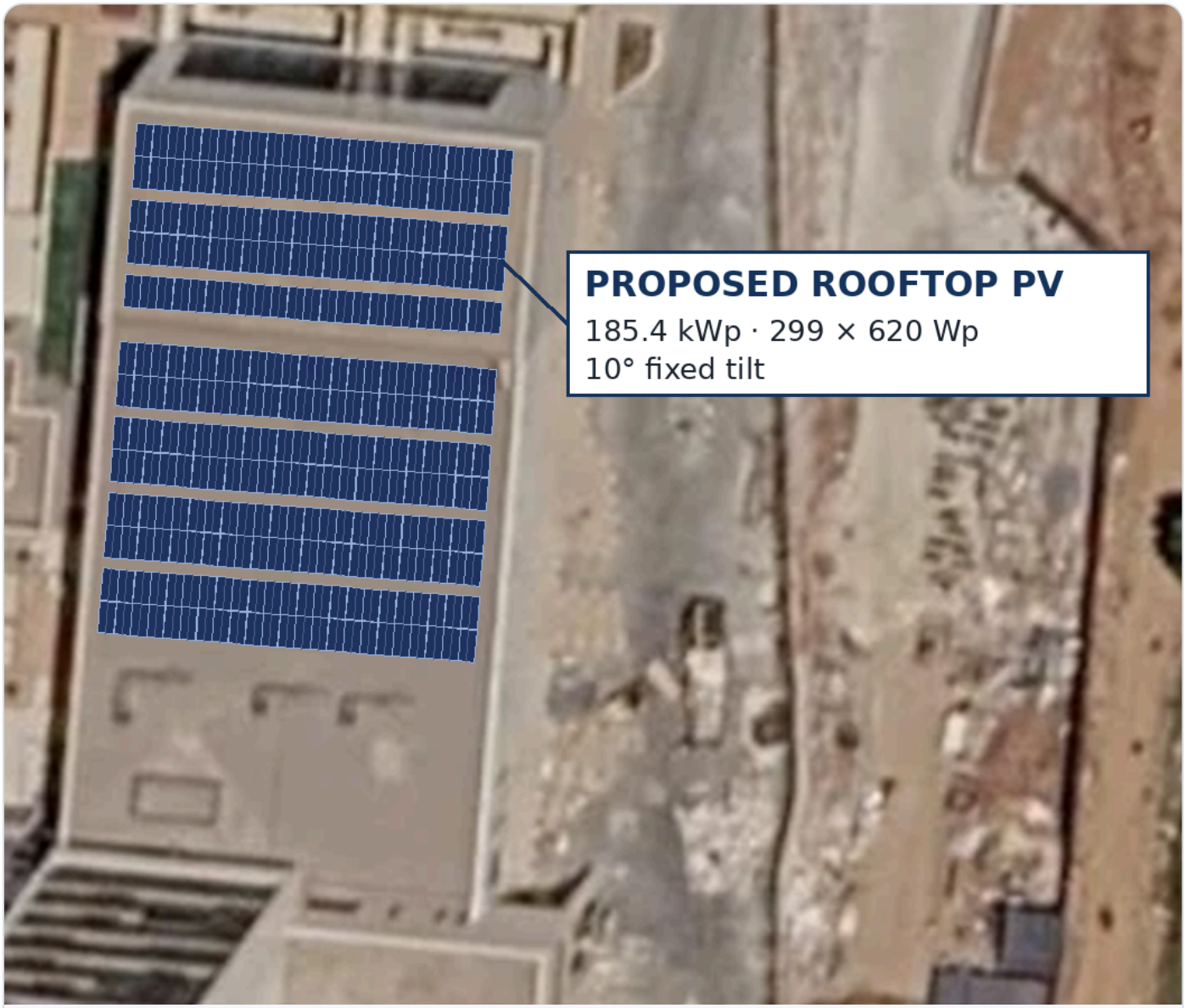
The Site

Kampala
Uganda
0°18'11.34"N,
32°35'13.76"E

Mezzanine
roof – architect's
solar zone only

≈ 1,000,000
kWh per month
hotel consumption

Equatorial
sun gives stable
output all year



Proposed PV array on the hotel's mezzanine roof (indicative visual)

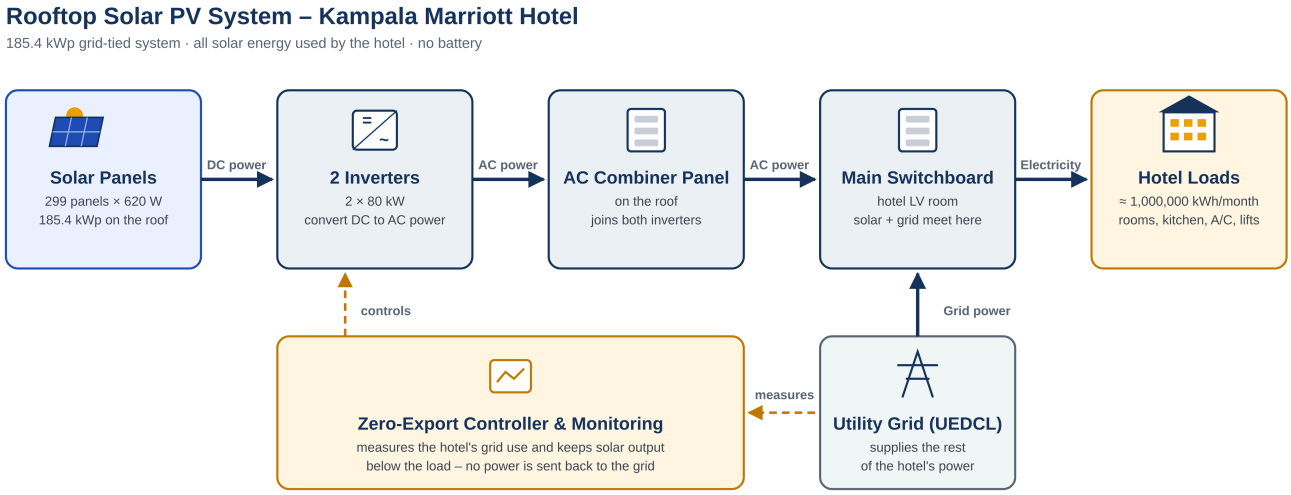
Proposed System

299
solar panels ×
620 W bifacial TOPCon

2 × 80 kW
string inverters

10° tilt
ballasted aluminium
(no roof penetration)

No battery
zero-export control
& smart monitoring



How the system connects – solar panels → inverters → AC combiner → main switchboard → hotel loads

Item	Specification
DC capacity	185.4 kWp (299 × 620 Wp, 2382 × 1134 × 30 mm)
AC capacity	160 kW (2 × 80 kW, 1100 V, multi-MPPT)
Connection	AC combiner panel on roof → spare 250 A way on main LV switchboard

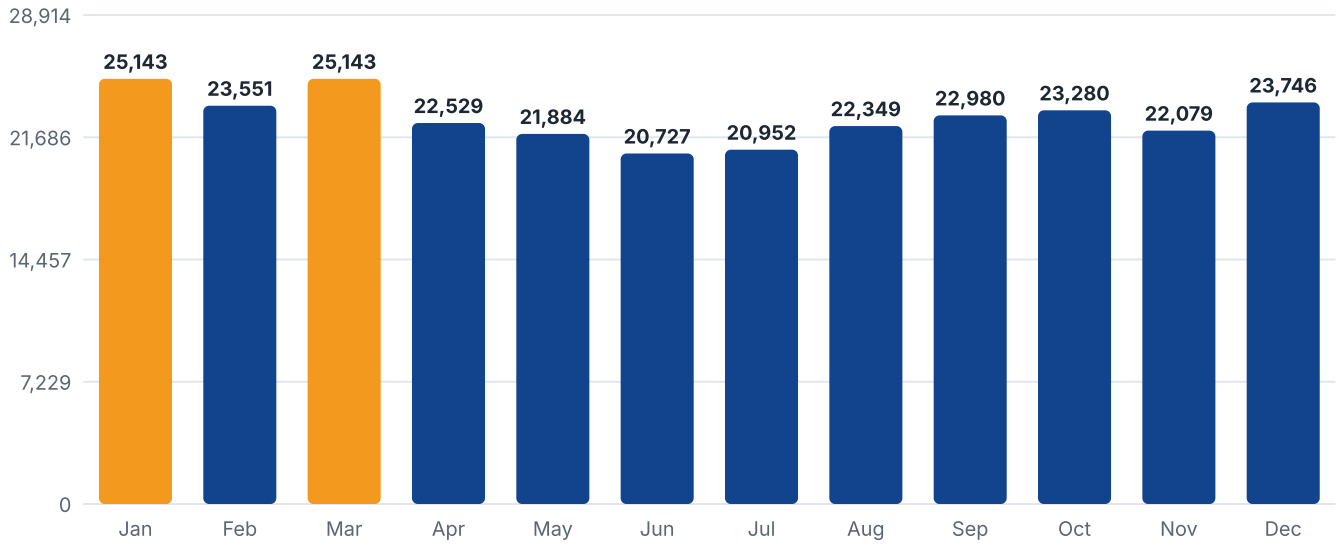
Energy Generation

274,362
kWh in year 1

752
kWh per day (average)

1,480
kWh per kWp per year (specific yield)

Monthly generation – year 1 (kWh)



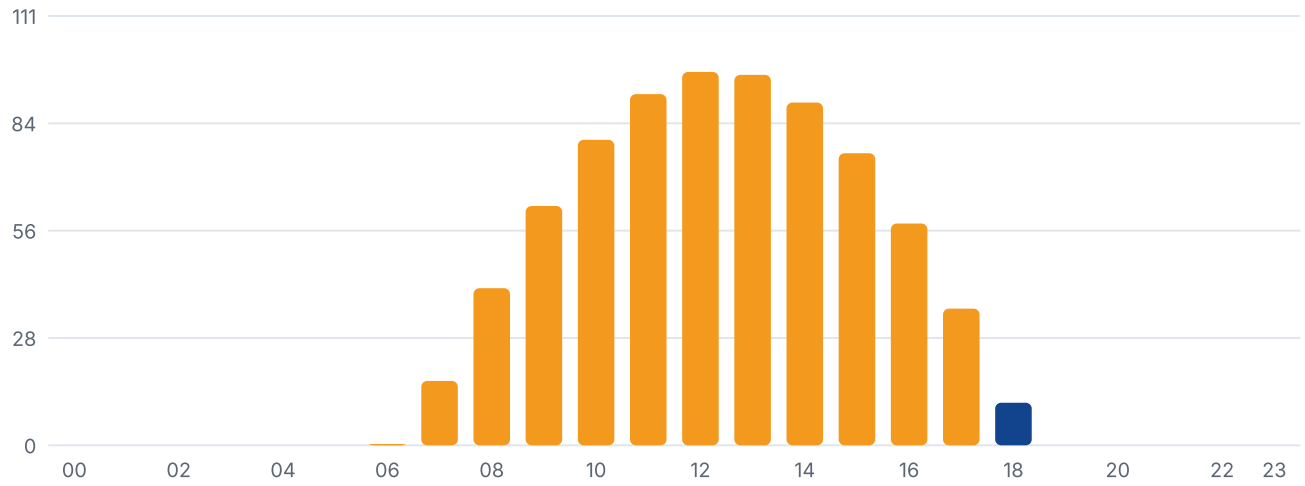
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
kWh	25.1k	23.6k	25.1k	22.5k	21.9k	20.7k	21.0k	22.3k	23.0k	23.3k	22.1k	23.7k
GHI	5.4	5.6	5.4	5.0	4.7	4.6	4.5	4.8	5.1	5.0	4.9	5.1

GHI = global horizontal irradiation, kWh/m²/day (typical Kampala values). Output degrades about 0.4 % per year; 25-year total ≈ 6.48 GWh.

Solar Energy Profile

Solar power is produced during daytime hours, exactly when the hotel's air-conditioning, kitchens and laundry load is high. Almost all solar energy falls in the **shoulder period (06:00–18:00)**.

Average daily solar output by hour (kW)



TOU band	Hours	Solar energy (kWh/yr)	Share of solar
Off-peak	00:00–06:00	0	0.0 %
Shoulder	06:00–18:00	270,333	98.5 %
Peak	18:00–24:00	4,030	1.5 %

98.5 %
of solar energy used in the shoulder band

2.3 %
of the hotel's annual consumption covered

0 kWh
exported to the grid (zero-export control)

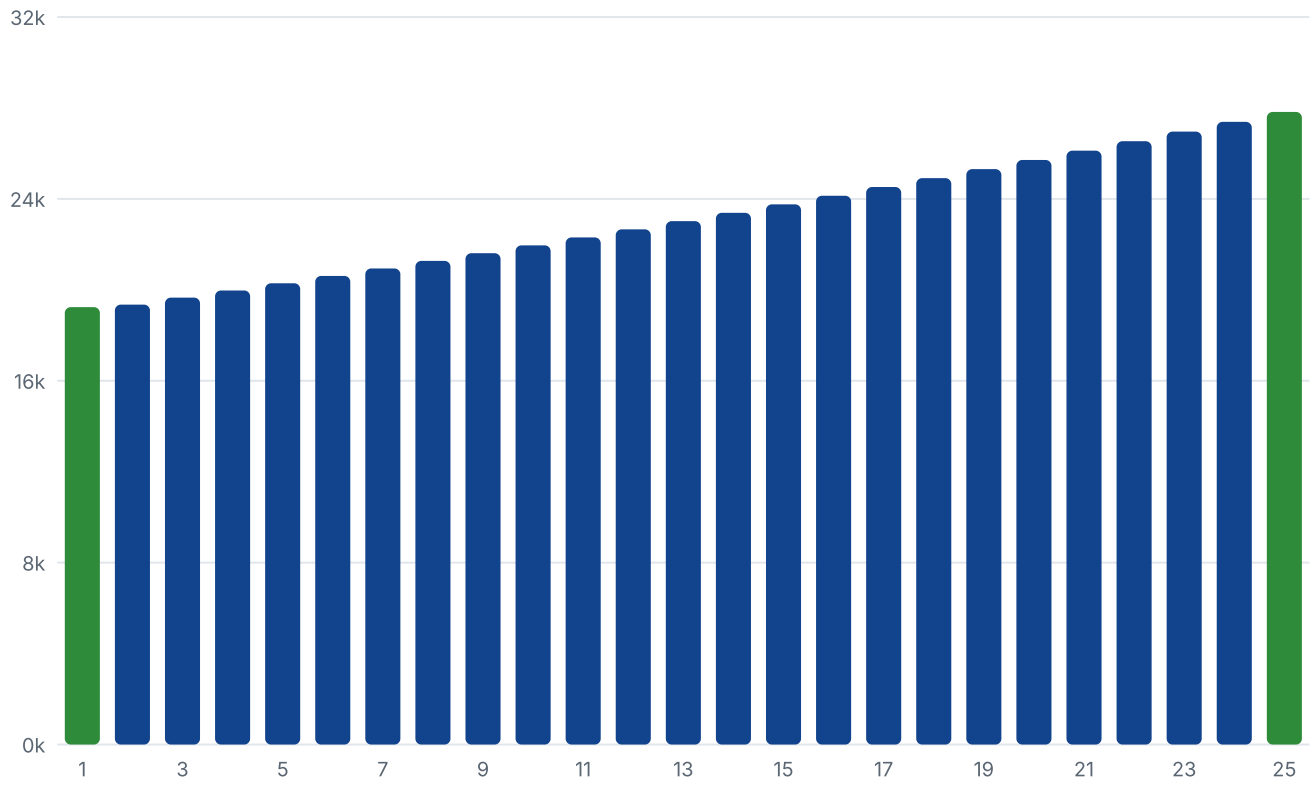
Savings

1
USD 19,212
year-1 savings
at USD 0.070/kWh

25
USD 27,786
year-25 savings
tariff rising +2 %/yr

Σ
USD 578,643
total savings
over 25 years

Annual electricity savings (USD)



Savings = solar energy used on site × grid tariff (USD 0.07002/kWh in year 1, escalating 2 % per year), after 0.4 %/yr module degradation. No O&M cost assumed; one inverter replacement (USD 9,600) in year 12.

Return on Investment

370%

ROI over 25 years

4.7× return

Every USD 1 invested returns about USD 4.7 in savings over the system life.

1	⟳	%
15.9 %	**6.1 years**	**16.6 %**
year-1 return	simple payback	IRR (25 years)

Metric	Value
Total project investment	USD 121,115
Net savings over 25 years (after inverter replacement)	USD 569,043
Net gain after recovering the investment	USD 447,928
Return on investment (net gain ÷ investment)	370 %

Financial Analysis

USD 121,115
investment (excl. VAT)

USD 71,221
NPV at 10 % discount

USD 0.052
per kWh solar cost
vs USD 0.070 grid

Parameter	Value
System size	185.4 kWp DC / 160 kW AC
Year-1 generation	274,362 kWh
Grid tariff (year 1) / escalation	USD 0.07002/kWh / 2 % per year
Module degradation	≈ 0.4 % per year
Year-1 savings	USD 19,212
Simple payback	6.1 years
IRR (25 years)	16.6 %
NPV (10 % discount rate)	USD 71,221
Levelised cost of energy (LCOE)	USD 0.052/kWh
O&M cost	Not included (none assumed)
Inverter replacement	USD 9,600 in year 12 (included)
25-year gross savings / net savings	USD 578,643 / USD 569,043

Cash Flow

Cumulative cash flow over the 25-year project life (USD). The investment is fully recovered during year 7.



6.1 yrs
payback – positive from year 7

+447,928
cumulative cash at year 25 (USD)

25 yrs
project life

Year	0	1	3	5	6	7	10	12	15	20	25
Annual	-121.1k	+19.2k	+19.6k	+20.3k	+20.6k	+20.9k	+21.9k	+13.0k	+23.7k	+25.7k	+27.8k
Cumulative	-121.1k	-101.9k	-63.0k	-22.7k	-2.2k	+18.7k	+83.5k	+118.8k	+188.9k	+313.3k	+447.9k

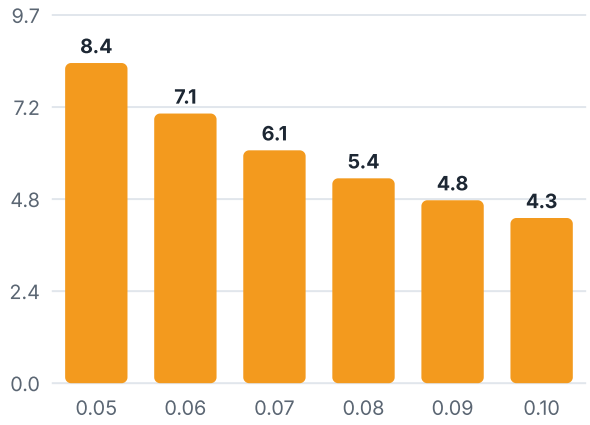
Year 12 includes the inverter replacement (USD 9,600).

Tariff Sensitivity

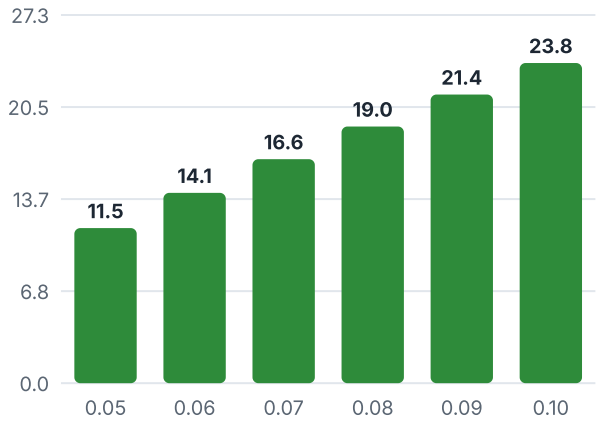
The higher the grid tariff, the better the returns. Even at USD 0.05/kWh the project still pays back and earns a positive NPV.

Tariff (/kWh)	Year-1 savings	Payback (yrs)	IRR	NPV @10 %
USD 0.05	USD 13,718	8.4	11.5 %	USD 15,349
USD 0.06	USD 16,462	7.1	14.1 %	USD 43,253
USD 0.07	**USD 19,205**	**6.1**	**16.6 %**	**USD 71,158**
USD 0.08	USD 21,949	5.4	19.0 %	USD 99,062
USD 0.09	USD 24,693	4.8	21.4 %	USD 126,967
USD 0.10	USD 27,436	4.3	23.8 %	USD 154,871

Payback (years)



IRR (%)



Break-even tariff ≈ **USD 0.0445/kWh** – below this tariff the NPV at 10 % would turn negative. Current tariff: USD 0.070/kWh.

What's Included & Warranties

WHAT'S INCLUDED

- Engineering & design
- 299 solar panels
- 2 inverters
- Mounting structure
- DC & AC cabling
- Monitoring & zero-export control
- Earthing & lightning protection
- Installation & commissioning
- Transport & customs clearing

WARRANTIES

12 / 30 yrs panels – product / linear power	**5 yrs** inverters (extendable)	**10 yrs** mounting structure	**2 yrs** workmanship from commissioning

Excluded: roof strengthening or waterproofing, battery storage, utility/permit fees, VAT and statutory levies, and works not listed above.

Commercial Quotation

Quotation No. Q-KMH-PV-2026-001	**Date** 05-11-2026	**Valid** 30 days (05-12-2026)
Client Kampala Marriott Hotel	**Supplier** Nile Sun Power Limited, Kampala, Uganda	

#	Scope of supply & works (turnkey EPC)	Qty
1	PV modules – 620 Wp bifacial N-type TOPCon (Tier-1)	299 nos
2	String inverters – 80 kW, 1100 V, multi-MPPT	2 nos
3	Ballasted aluminium mounting structure, 10° tilt	1 lot
4	DC system – PV cable, connectors, string isolators, cable tray	1 lot
5	AC system – AC combiner panel, MCCBs, cabling to main LV switchboard	1 lot
6	Smart data logger, remote monitoring & zero-export control	1 set
7	Earthing & lightning protection integration	1 lot
8	Installation, testing & commissioning	1 lot
9	Transport, logistics & customs clearing to site	1 lot
TOTAL TURNKEY EPC PRICE (excluding VAT)		**USD 121,115**

Equivalent to USD 0.65 per Wp · One hundred twenty-one thousand one hundred fifteen US Dollars only.

Commercial terms	
Payment	30 % advance with PO · 60 % on delivery of main equipment · 10 % on commissioning
Delivery	Equipment 8–10 weeks from PO; installation & commissioning ≈ 4 weeks (≈ 12–14 weeks total)
Taxes	VAT, withholding tax and statutory levies excluded – charged as applicable
Client scope	Roof access, spare 250 A way on the main LV switchboard, power & water during installation

For Nile Sun Power Limited

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Signature / Name / Date

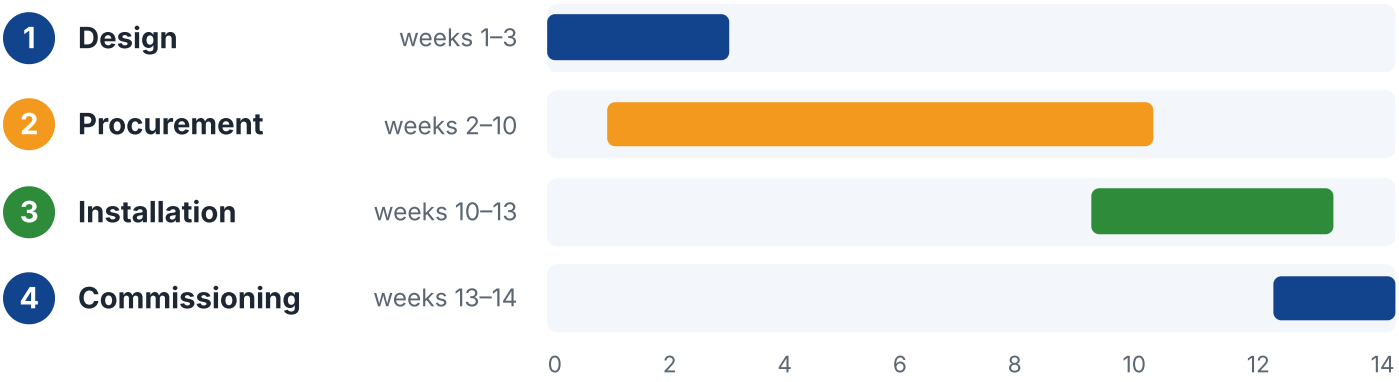
Accepted by Kampala Marriott Hotel

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Signature / Name / Date

Timeline & Contact

Project timeline (≈ 14 weeks from purchase order)



Thank you

We appreciate the opportunity to support Kampala Marriott Hotel with a clean, reliable and sustainable energy future.

COMPANY
Nile Sun Power Limited
Kampala, Uganda

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